

PROBLEM SPACE

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

Setting A — HCI: Mercer ED Triage Desk

USER

Group(s)	Characteristics	Needs
ED triage nurses responsible for assessing ^{ting} patients, recording clinical information, assigning an Emergency Severity Index (ESI) level, and monitoring patients awaiting reassessment. primary users secondary users Emergency physicians, charge nurses, bed managers, Observation Unit staff, clinical IT staff and hospital governance personnel who review triage decisions, alerts or system performance. Patients using the waiting portal to report symptom changes.	Clinically trained staff working under significant time pressure. They may assess several patients in rapid succession while recording vital signs, asking questions, managing queues and responding to interruptions. Users may also be affected by fatigue during long or overnight shifts and may have different levels of confidence with digital systems. Nurses must also monitor patient deterioration after initial triage while managing competing clinical priorities. These users have different clinical, operational and technical responsibilities. Clinicians require clear and relevant patient information, while IT and governance staff require reliable records, system transparency and evidence that the tool is operating safely. Many users require different levels of access, ranging from clinical decision-making to governance, audit and system administration.	• Rapid access to reliable AI-assisted triage recommendations. • Clear explanations supporting each recommendation. • A quick method to confirm or override AI suggestions. • Immediate visibility of patient deterioration alerts submitted through the patient portal. • Reliable operation during busy clinical shifts. • Minimal disruption to existing triage workflows. • Secure audit trails of recommendations, overrides and reassessments. • Reliable system performance data for governance and quality improvement. • Transparency into how AI recommendations were generated. • Evidence supporting safe clinical deployment and future validation.
• Complete patient triage efficiently without delaying patient care. • Receive an AI-assisted ESI recommendation that supports rapid clinical decision-making. • Confirm or override the recommendation while maintaining full clinical control. primary users secondary users • Review AI recommendations and nurse overrides for quality assurance. • Monitor system performance and identify workflow issues during pilot deployment. SHORT-TERM	Goal(s) • Improve consistency and accuracy of triage decisions across different shifts and staff. • Reduce cognitive workload while maintaining patient safety. • Increase confidence in AI-assisted clinical decision support without creating dependence on the system. • Improve communication between waiting patients and clinical staff through structured symptom reporting. • Improve consistency and accuracy of triage decisions across different shifts and staff. • Reduce cognitive workload while maintaining patient safety. • Increase confidence in AI-assisted clinical decision support without creating dependence on the system. LONG-TERM	

AI TRIAGE SUPPORT SYSTEM

Task(s)			
• Receive patient information from manual entry. • Analyse clinical features using the trained triage model. • Predict the recommended ESI level. • Display confidence level and contributing clinical factors. • Allow the nurse to confirm or override the recommendation. • Receive and review patient-reported symptom updates during the waiting period. SHORT-TERM	Advantages What tasks does the AI system perform? • Simple interface that is easy to interpret under pressure. • Reduces cognitive workload during busy shifts.		• Improve consistency of triage decisions. • Reduce cognitive workload during busy shifts. • Maintain a complete audit trail of recommendations and overrides. • Support continuous improvement of the model using reviewed clinical data. • Support reassessment of deteriorating patients using updated clinical information. LONG-TERM
• Provides rapid clinical decision support. • Highlights patients requiring reassessment based on AI recommendations, low-confidence predictions and patient-reported symptom updates. • Reduces variation in triage recommendations. Clinical Support	• Fits naturally into the existing triage workflow. • Does not interrupt patient assessment. • Supports communication between waiting patients and the triage team without replacing face-to-face assessment. User Experience	• Displays confidence score. • Shows the key factors influencing the prediction. • Supports transparent decision-making. Explainability	• Produces recommendations within seconds. • Allows immediate nurse confirmation or override. • Logs all clinical decisions. Decision Support
• Accepts manually entered patient information. • Can support future EHR integration. Data Integration	• Manual workflow remains available if the system fails. • Low-confidence predictions prompt additional clinical review. Reliability	• Can connect to Mercer EHR. • Supports audit logs and reporting. • Stores prediction history for governance purposes. • Integrates patient portal updates into the nurse dashboard. Connection to system	